

Leads Scoring for X Education - Case Study

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PROBLEM STATEMENT

- X Education is an education company which sells online courses to industry professionals.
- Whenever people land on its website, they might browse the courses, fill out a form for the course, or watch some videos. When these people fill out a form with their email address or phone number, they are classified as leads.
- Once these leads are acquired, employees from the Sales team start making calls, writing emails, etc. Through this process, some of the leads get converted into successful sales. The typical lead to successful sale conversion rate at X Education is around 30%.
- The company wishes to identify the most potential leads, also known as Hot Leads. If they successfully identify this set of leads, the lead conversion rate would go up as the Sales team would now be focusing more on communicating with the potential leads rather than making calls to everyone.

OBJECTIVE

- Build a Logistic Regression Model to assign a lead score between 0 and 100 to each of the leads, which can be used by the company to target potential leads. A higher score means that the lead is hot, i.e., most likely to convert, while a lower score would mean that the lead is cold and will mostly not get converted.

SOLUTION STEPS

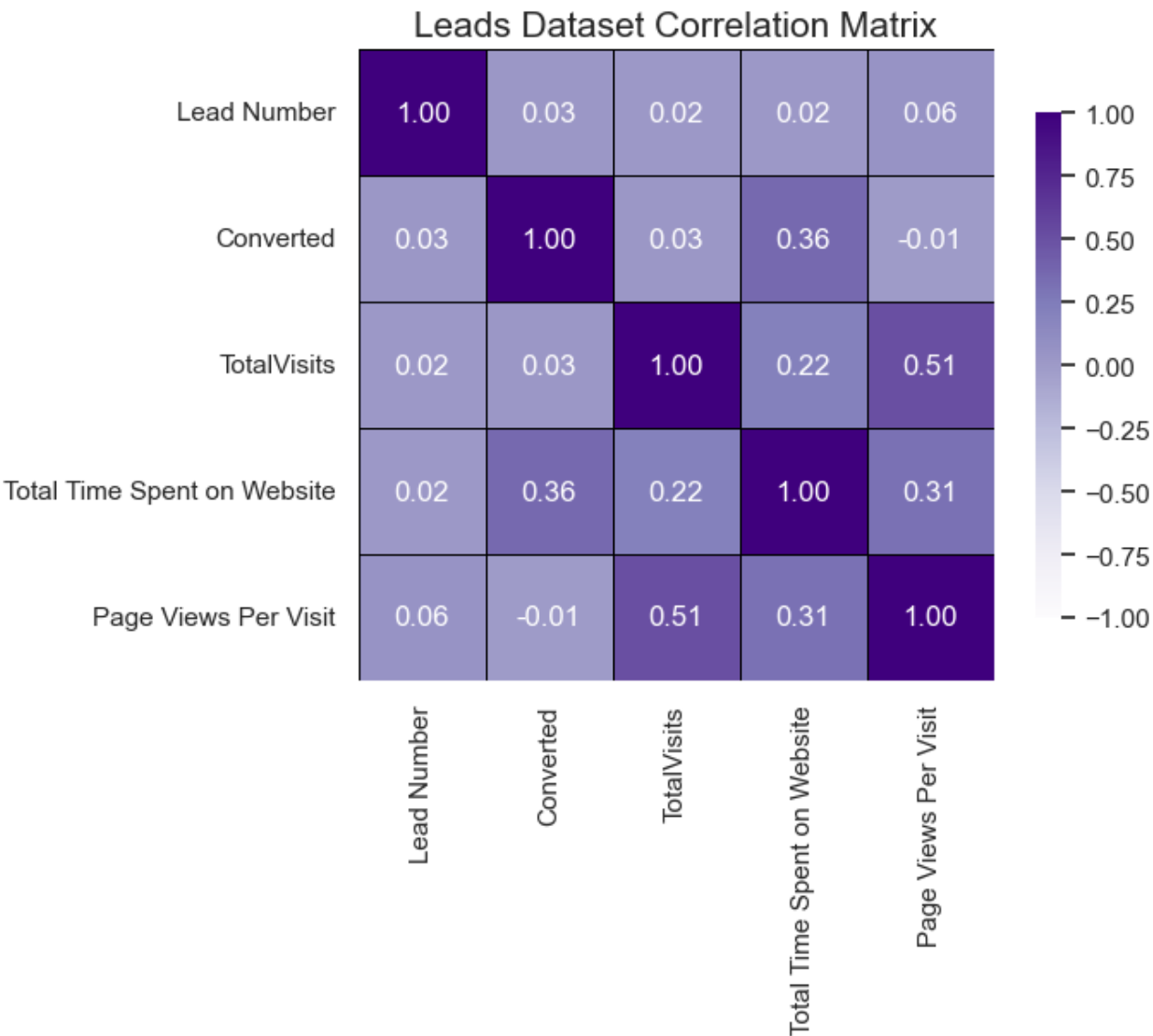
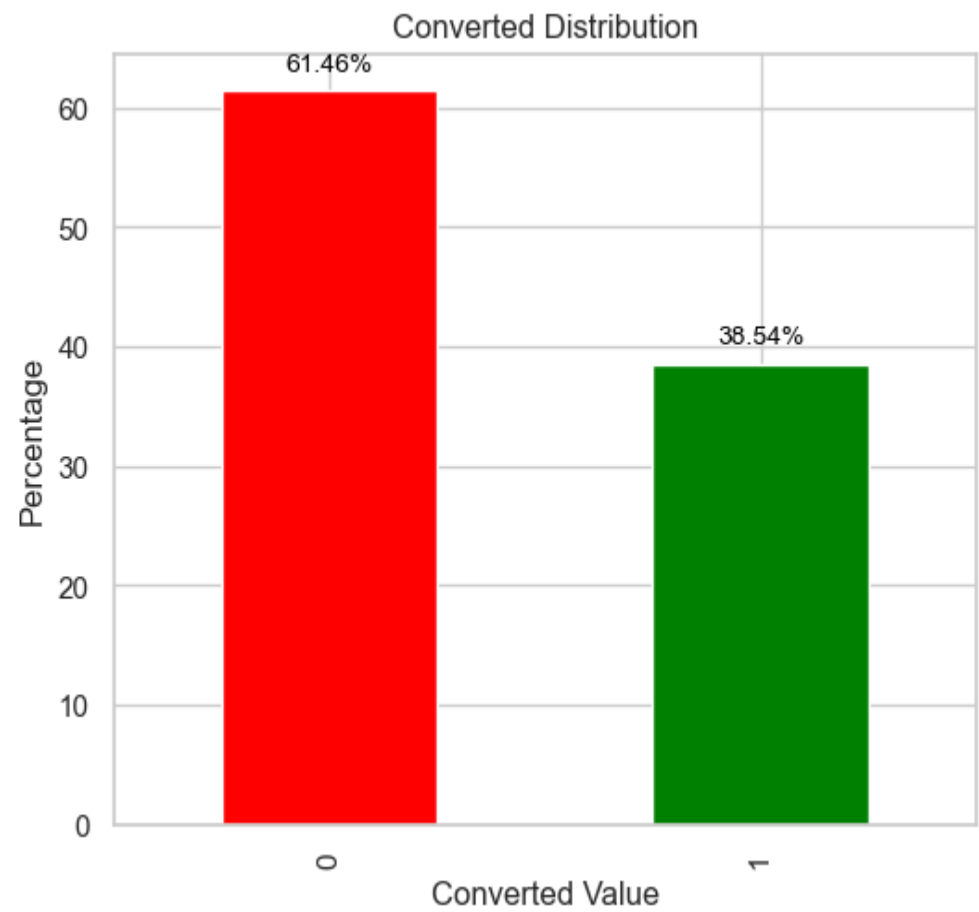
- Data cleaning and data manipulation
 - Check and handle duplicate data
 - Check and handle missing values
 - Check and handle data inconsistencies
 - Check and handle outliers
- EDA
 - Univariate data analyses
 - Bivariate data analyses
- Dummy Variables & Data Preparation
- Train-Test Split
- Feature Scaling & Model Building
- Model Evaluation & Predictions
- Conclusions & Recommendations

DATA CLEANING

- The original dataset has 9240 rows and 37 columns.
- It was only partially clean, and had missing values and inconsistencies in data formats.
- Columns with null values greater than 30% were dropped.
- 'Select' level present in few columns was replaced with null value.
- Null values in other columns were imputed with either median or most frequently occurring value.
- No duplicates or outliers were present.
- Inconsistency in data formats was corrected.
- Other unimportant columns were dropped.

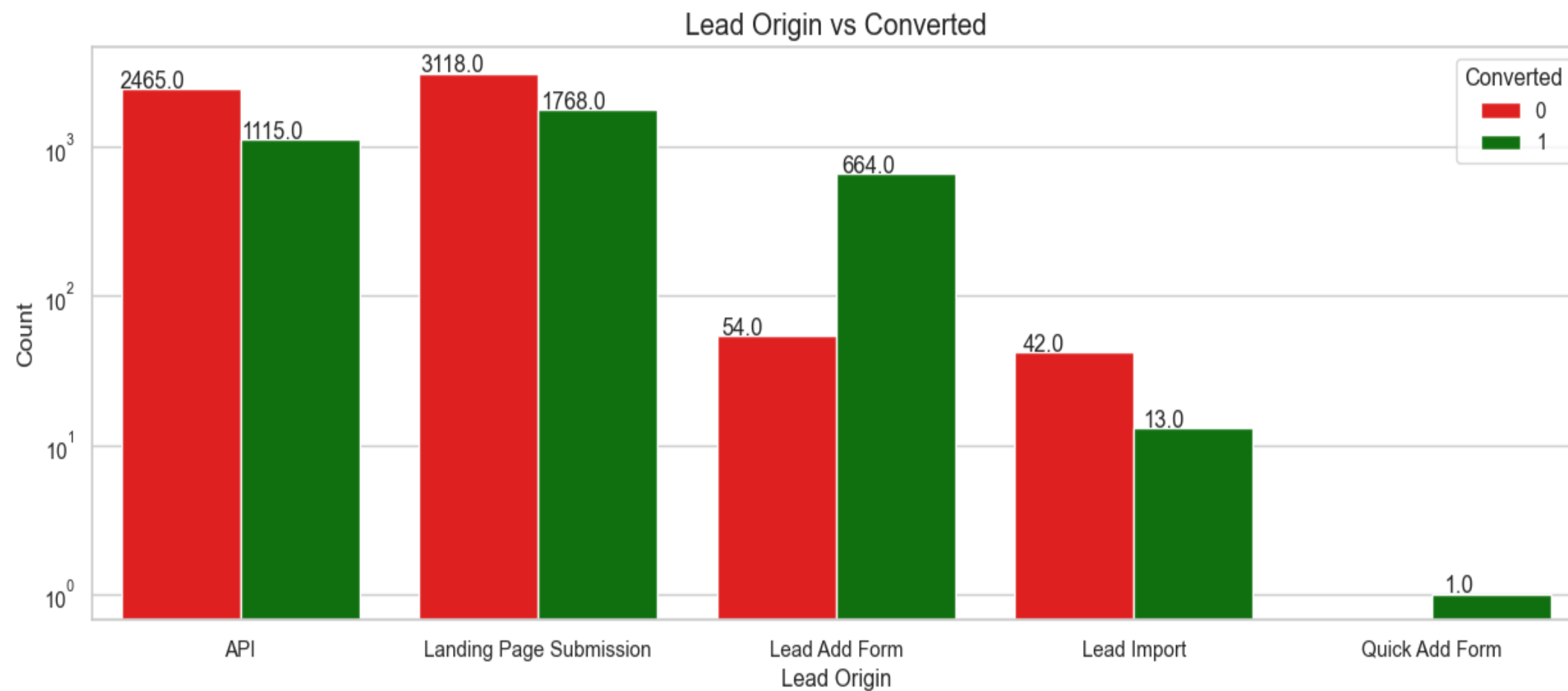
EDA

Distribution of target variable

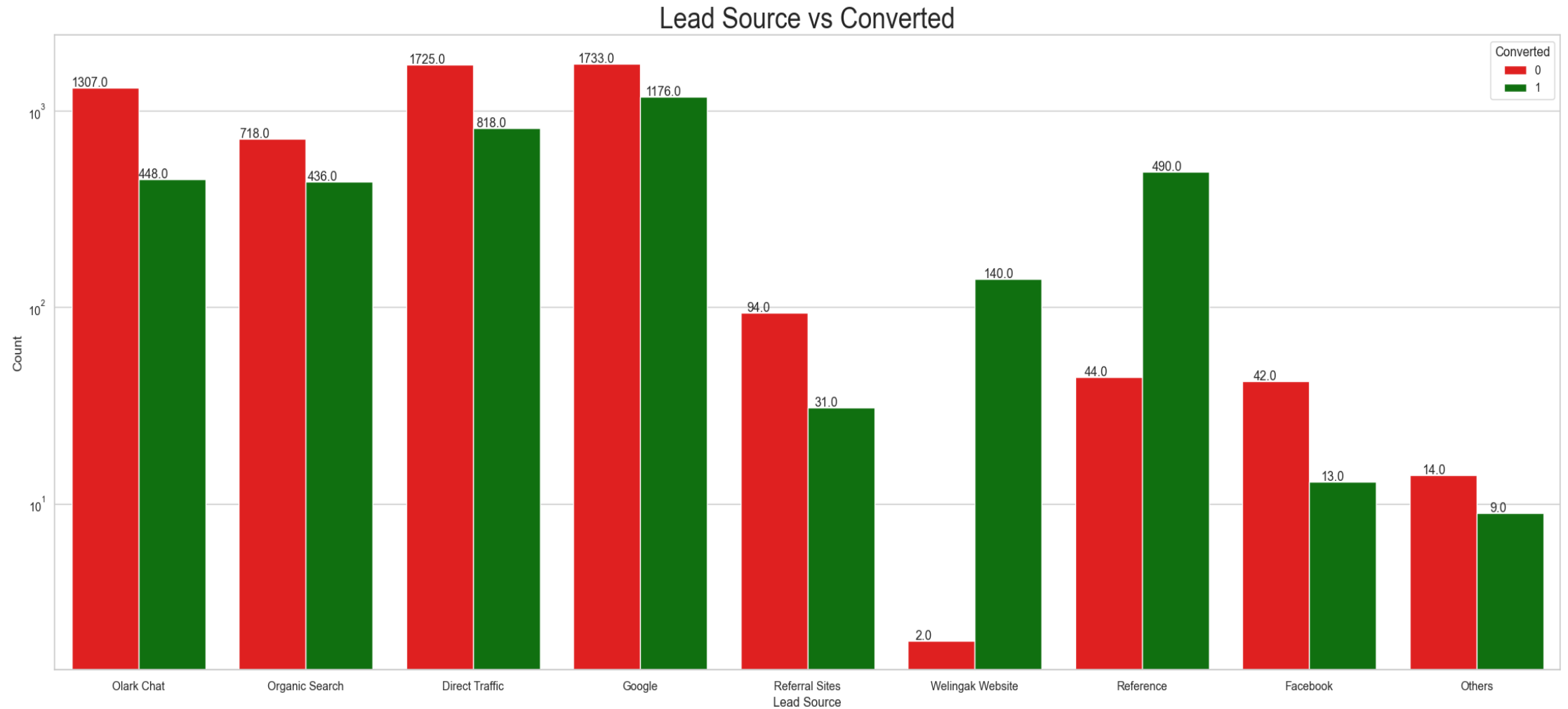


Correlation Matrix of numerical columns in the dataset

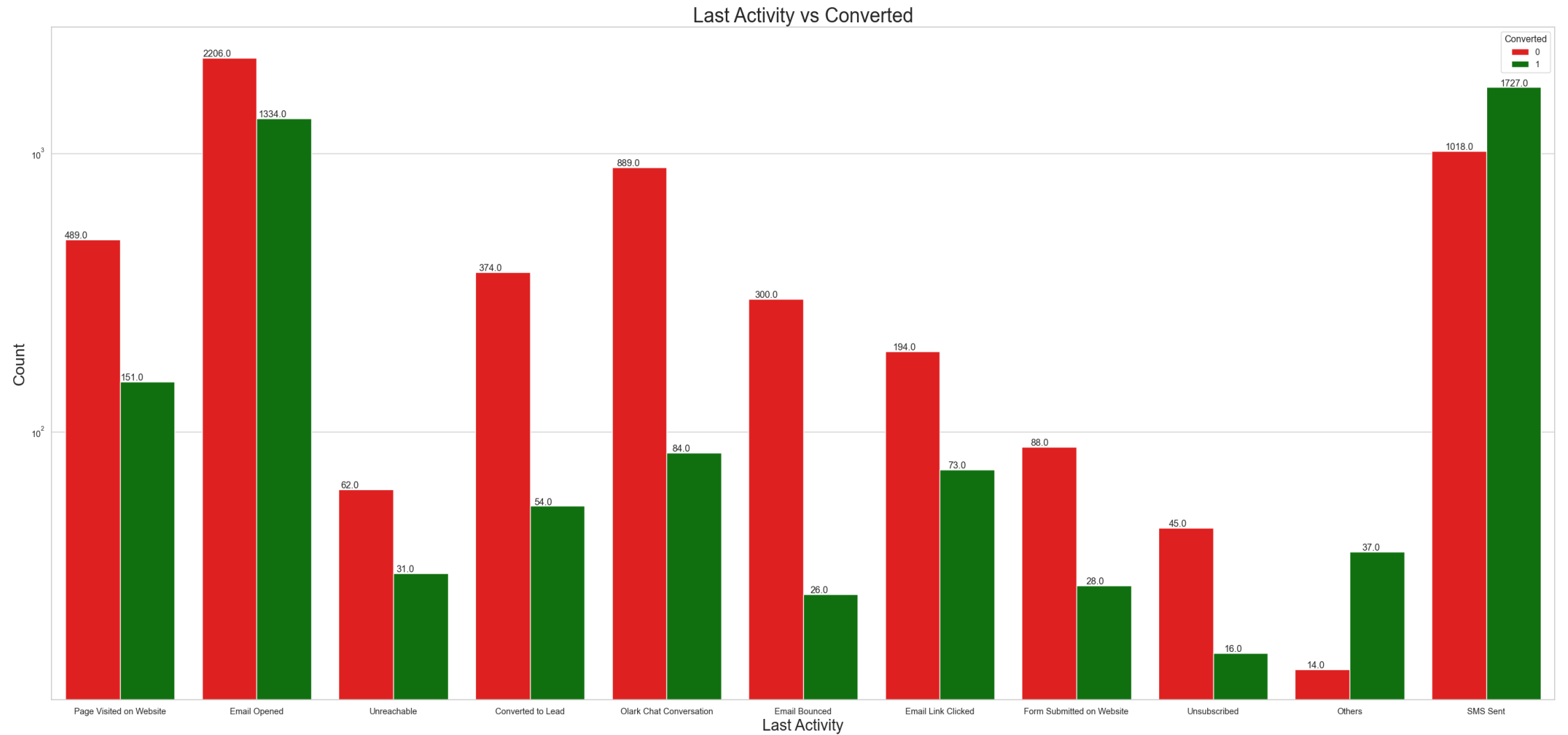
'Lead Origin' vs 'Converted'



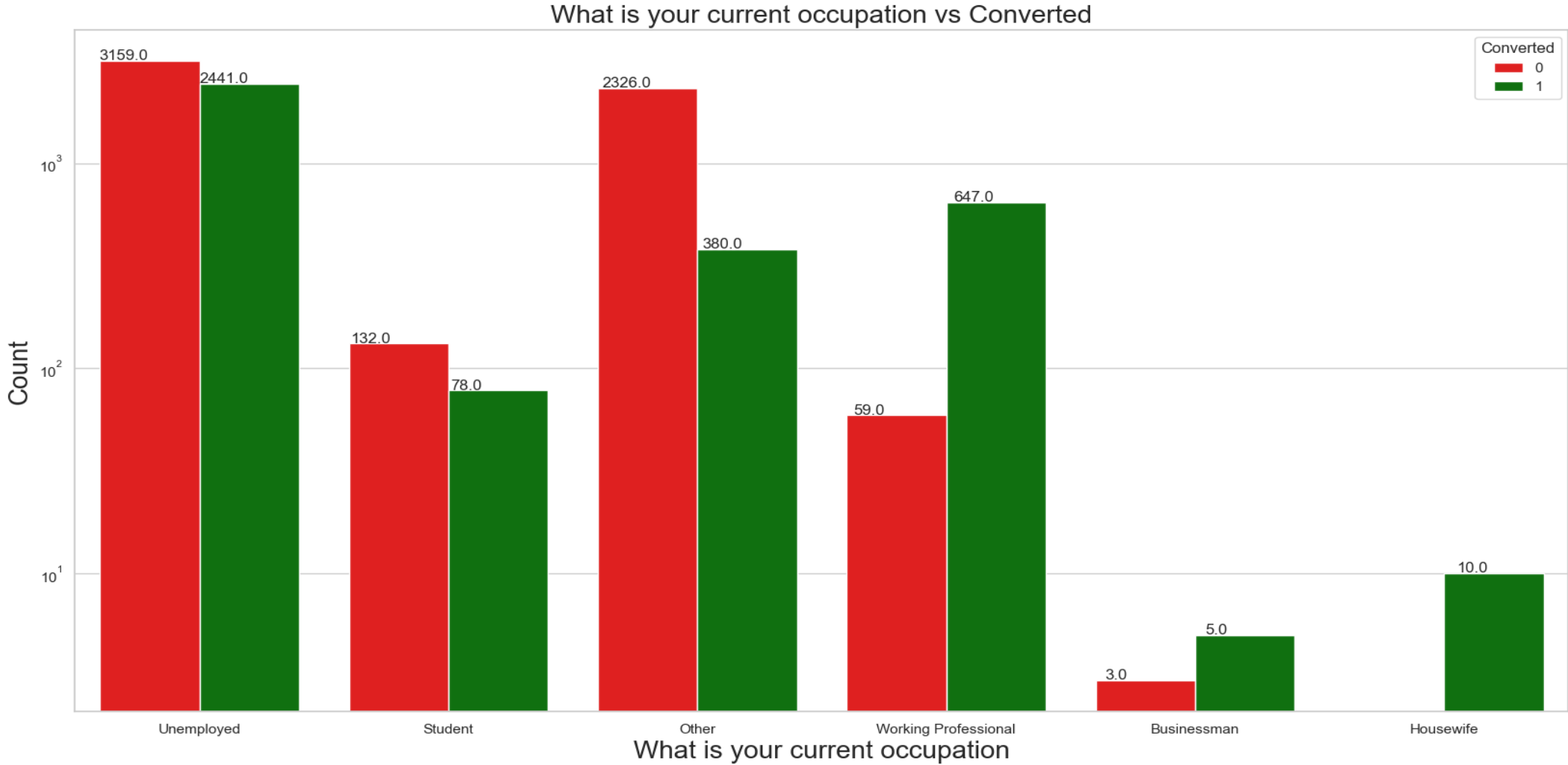
'Lead Source' vs 'Converted'



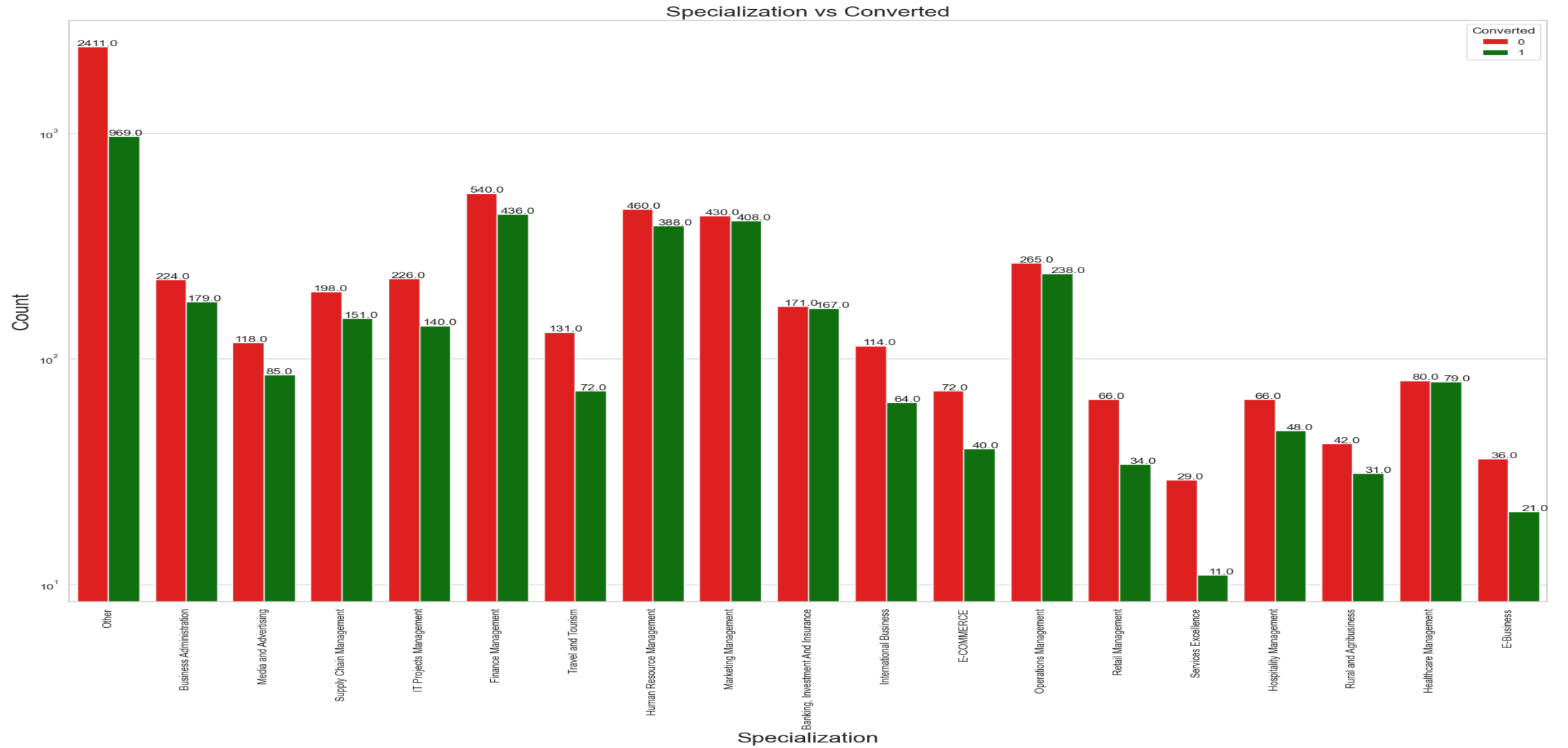
'Last Activity' vs 'Converted'



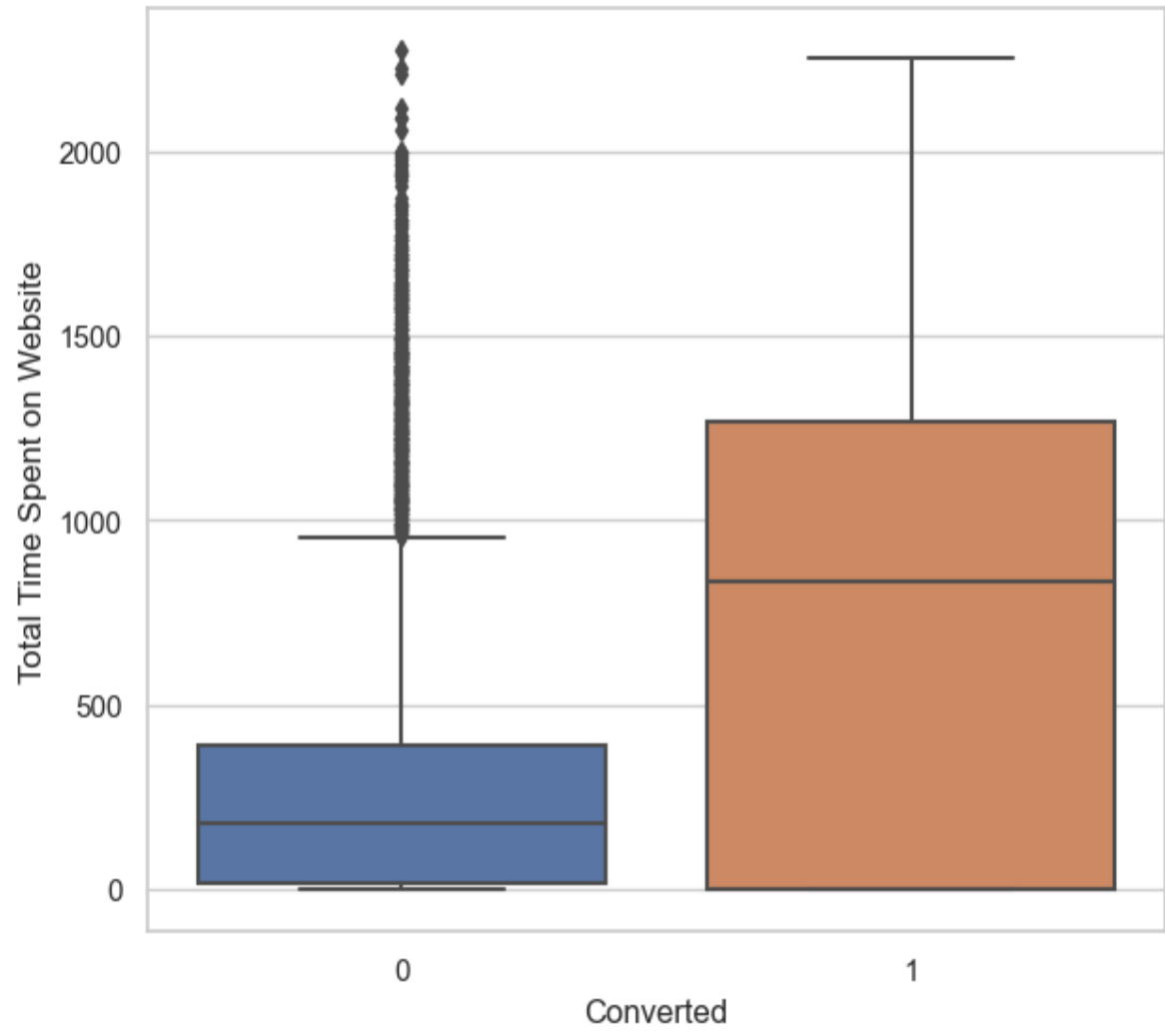
‘What is your current occupation’ vs ‘Converted’



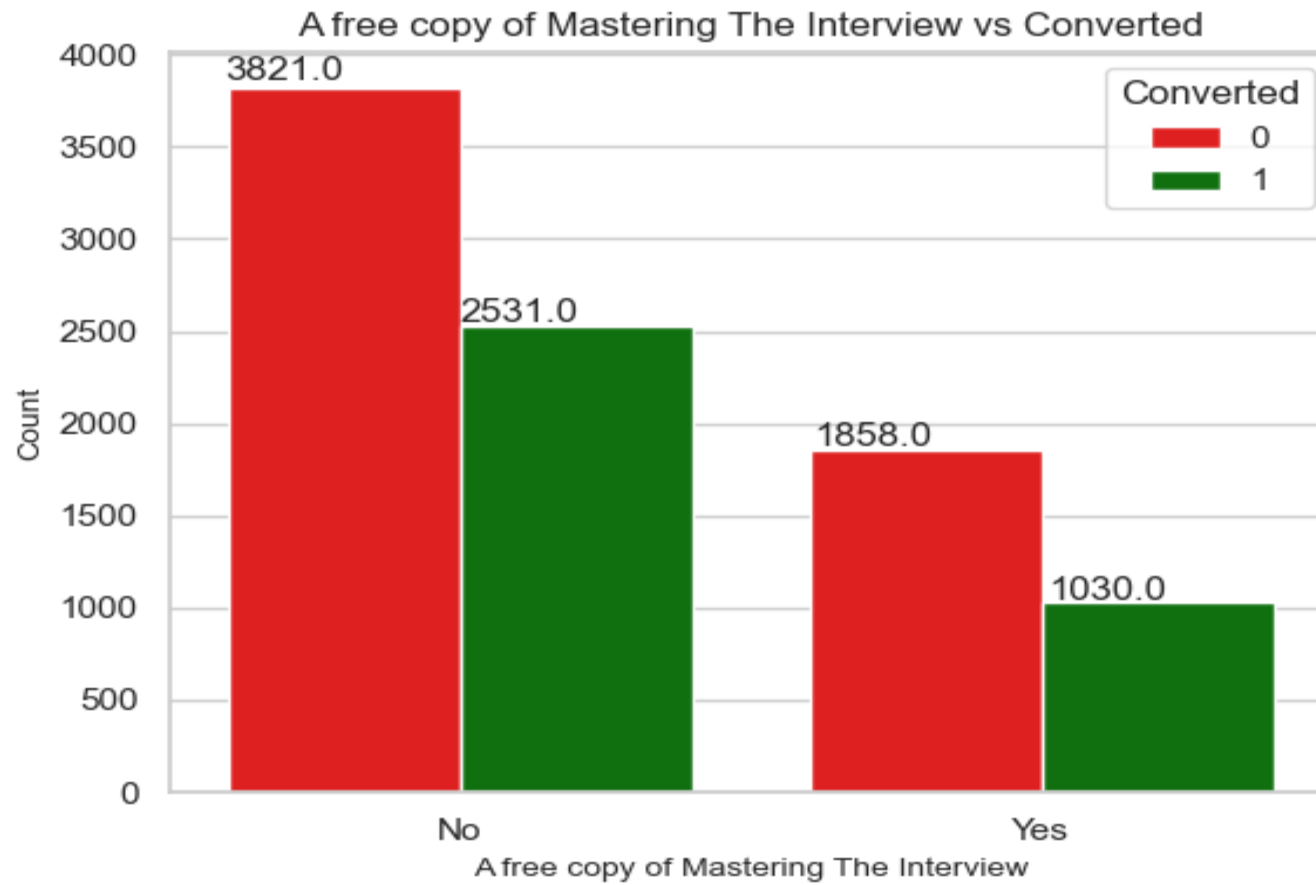
'Specialization' vs 'Converted'



'Total Time Spent on Website' vs 'Converted'



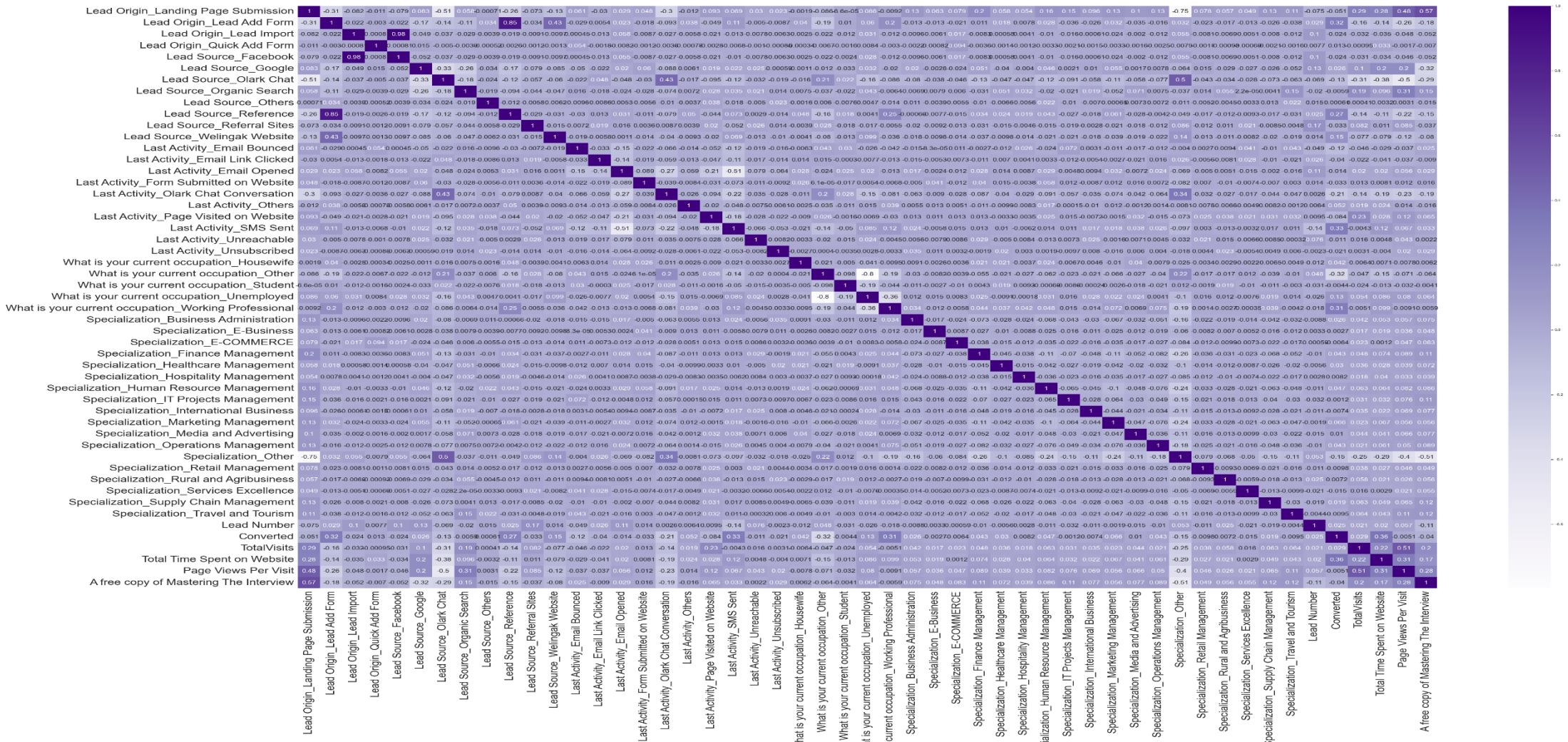
'A free copy of Mastering The Interview' vs 'Converted'



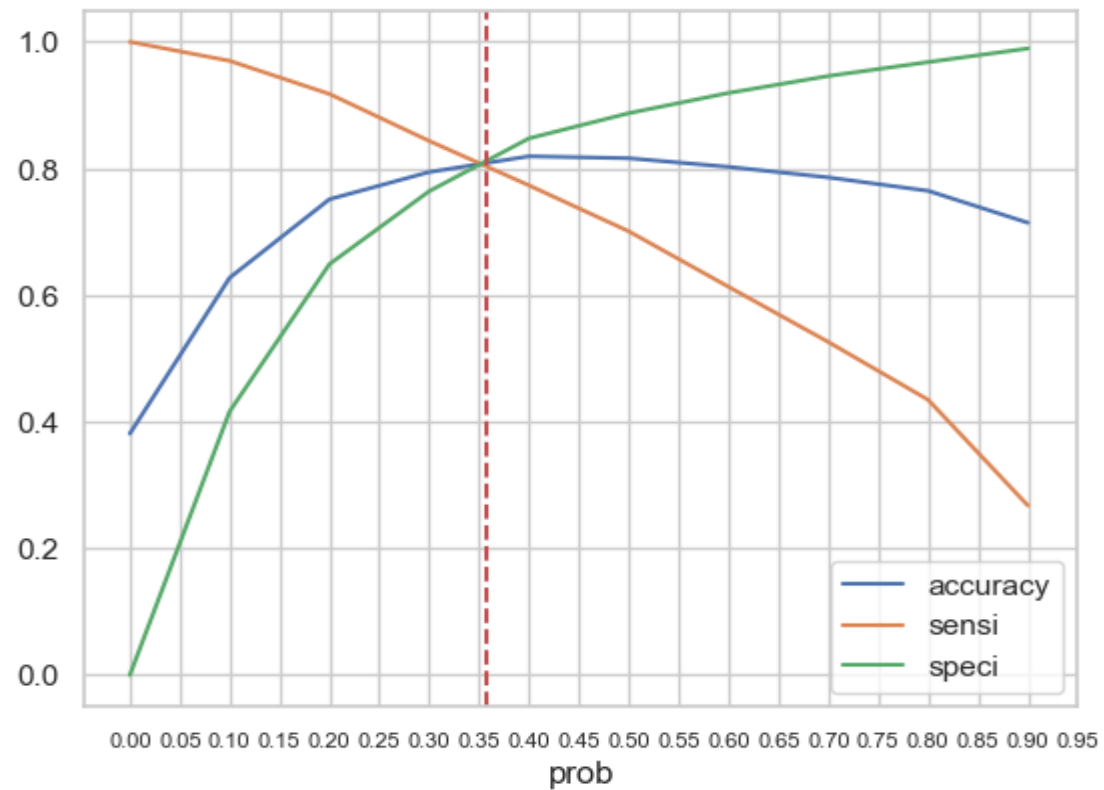
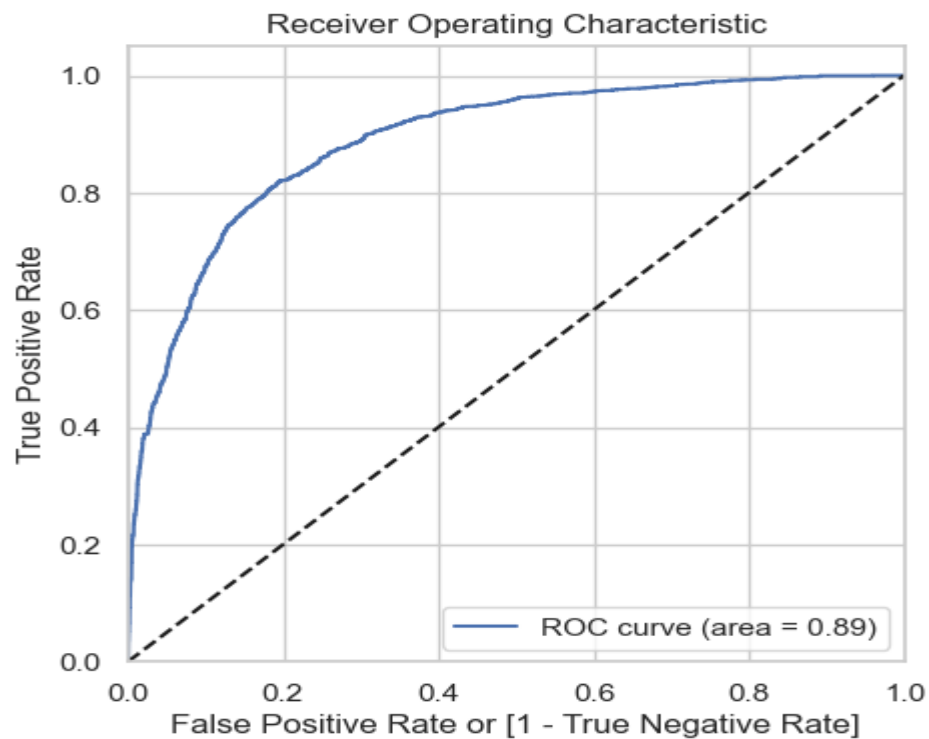
MODEL BUILDING

- Splitting the data into Training and Testing datasets
- Basic step for logistic regression is to perform a train-test split, we have chosen 70:30 ratio
- Use RFE for Feature Selection
- Running RFE with 15 variables as output
- Building Model by removing variables whose p-value is greater than 0.05 and vif value is greater than 5
- Predictions on train and test datasets
- Overall accuracy 81%

Heatmap showing correlation coefficients between variables



ROC Curve



Optimal cutoff probability is at 0.358

Conclusion

The variables that matter the most are -

- Total time spent on the website
- When their current occupation is a working professional
- When the lead source was
 - a. Google
 - b. Direct Traffic
 - c. Organic search
 - d. Welingak website
- When the last activity was
 - a. SMS
 - b. Olark chat conversation
- When the lead origin is Lead Add Format

Keeping these in mind, X Education can flourish if they focus on leads satisfying the above criteria. They will be able to effectively increase their lead conversion rate by a significant number